

Differential Geometry on Signed Hypergraphs

A primer for the GömbSoma–Ricci–Stim architecture

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Abstract

This primer collects, in roughly the order they appear in the GömbSoma–Ricci–Stim plan, the differential-geometric machinery that makes “hypergraph convolution” a principled object rather than an ad-hoc message-passing rule. We cover: simplicial complexes and signed boundary operators; the Hodge Laplacian at each simplex dimension; the Hodge decomposition theorem; the heat kernel; Forman–Ricci and Ollivier–Ricci curvature; the Bochner–Weitzenböck identity that couples curvature to the Laplacian; and discrete Ricci flow as a graph-rewiring tool. Each section has a small example, a TikZ figure, and a one-line takeaway for how the concept enters the architecture.

1 Why bother

A standard CNN is a discretisation of the heat equation $\partial_t u = \Delta u$ on a planar grid: the convolution kernel is a finite-difference Laplacian, ReLU is a flow-rectifier, pooling is a local average. The whole CNN stack is a clumsy approximation of a continuous geometric process.

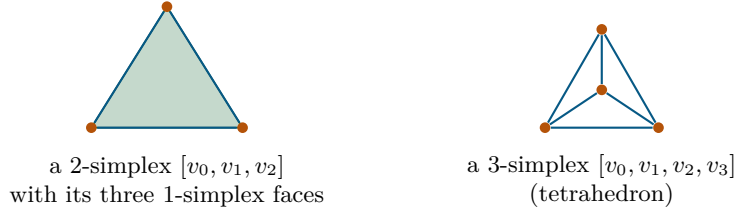
Hypergraph networks generalise this to irregular topology (V, E instead of an $H \times W$ grid). *Signed* hypergraphs generalise further: edges carry a sign $\sigma(e) \in \{-1, +1\}$ encoding polarity (trust / distrust, rotational / prismatic, brightness-up / brightness-down). The full Riemannian-geometric stack — boundary operators, Laplacians, curvature, heat flow, Bochner identities — transfers across these generalisations almost verbatim. This primer makes that transfer explicit.

The one-sentence takeaway: *the architecture we want to build is the natural discretisation of Riemannian geometry on a signed simplicial complex*. Forman gave us Ricci curvature [1]; Topping et al. gave us the over-squashing diagnosis [2]; Cartwright & Harary gave us the signed-balance theory [3]. The architecture is what happens when you put them in the same box.

2 Simplicial complexes

Definition 2.1 (Simplicial complex). *A finite collection K of finite subsets of a vertex set V , closed under taking subsets: if $\sigma \in K$ and $\tau \subseteq \sigma$, then $\tau \in K$. Elements of K are called simplices; a simplex of size $k + 1$ is a k -simplex.*

The 0-simplices are vertices; 1-simplices are edges; 2-simplices are (filled) triangles; 3-simplices are tetrahedra; and so on. A signed hypergraph $H = (V, E, \sigma)$ presents the data of a signed simplicial complex if we additionally include the triangles (and higher) determined by the σ -product structure.



Architectural use. In GömbSoma the 0-simplices are patches (or anchors), 1-simplices are signed edges (with $\sigma \in \{-1, +1\}$), 2-simplices are signed triangles (with $\pi(c_3) \in \{-1, +1\}$). Higher simplices correspond to longer cycles enumerated by the layer machinery.

3 Signed boundary operators

Definition 3.1 (Signed boundary ∂_k). *The boundary of an oriented k -simplex $\sigma = [v_0, v_1, \dots, v_k]$ is the formal alternating sum*

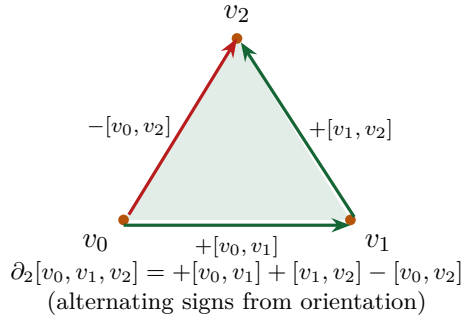
$$\partial_k \sigma = \sum_{i=0}^k (-1)^i [v_0, \dots, \hat{v}_i, \dots, v_k]$$

where $\hat{v}_i$ denotes omission. On a signed complex, each term is additionally multiplied by the simplex's σ -product polarity.

Example. For a triangle $[v_0, v_1, v_2]$:

$$\partial_2[v_0, v_1, v_2] = [v_1, v_2] - [v_0, v_2] + [v_0, v_1].$$

That is, the boundary is the sum of the three edges with alternating signs — consistent with the right-hand-rule orientation around the triangle.



3.1 The fundamental identity

Theorem 3.2 ($\partial \circ \partial = 0$). *For any k , $\partial_{k-1} \circ \partial_k = 0$. Equivalently, the image of ∂_k is contained in the kernel of ∂_{k-1} .*

This is the cornerstone of homology. The boundary of a boundary is zero: the perimeter of a triangle, viewed as a chain of edges, has zero net boundary because each vertex appears once with $+1$ and once with -1 . This identity is what makes the Hodge Laplacian well-defined.

Architectural use. In GömbSoma the signed boundary ∂_2 takes signed triangles to signed edges and is the algebraic backbone of the Cartwright–Harary balance gate: a triangle's σ -product is $+1$ iff its boundary (as a chain) is "trivial" in the appropriate quotient.

4 Hodge Laplacian

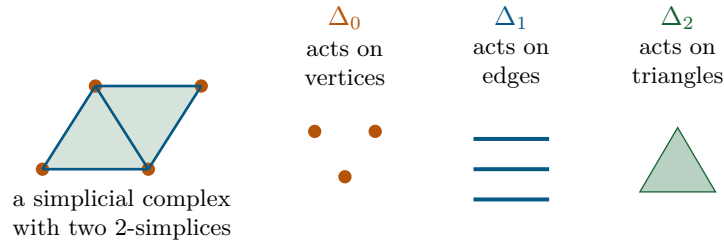
Definition 4.1 (Hodge Laplacian at dimension k).

$$\Delta_k = \partial_{k+1} \partial_{k+1}^* + \partial_k^* \partial_k$$

where ∂^* is the transpose (the coboundary or adjoint boundary operator).

The familiar graph Laplacian $\Delta_0 = D - A$ is precisely Δ_k at $k = 0$. The novelty at higher k :

- Δ_0 acts on *vertex* feature vectors. The standard graph Laplacian.
- Δ_1 acts on *edge* feature vectors. Captures edge-flow dynamics; vortices and gradients are visible in its spectrum.
- Δ_2 acts on *triangle* feature vectors. Captures face-flow; relevant when the data has well-defined 2-simplices.



4.1 Hodge decomposition

Theorem 4.2 (Hodge decomposition). *Any k -chain ω decomposes uniquely as*

$$\omega = \omega_{\text{harm}} + \partial_{k+1} \alpha + \partial_k^* \beta$$

where $\omega_{\text{harm}} \in \ker \Delta_k$ is harmonic, $\partial_{k+1} \alpha$ is a boundary (“curl-free” analogue), and $\partial_k^* \beta$ is a coboundary (“gradient” analogue).

The harmonic part is the topologically invariant content. Its dimension is the k -th Betti number β_k — a count of k -dimensional “holes” in the complex. For a connected graph, $\beta_0 = 1$ (one connected component); β_1 counts independent cycles.

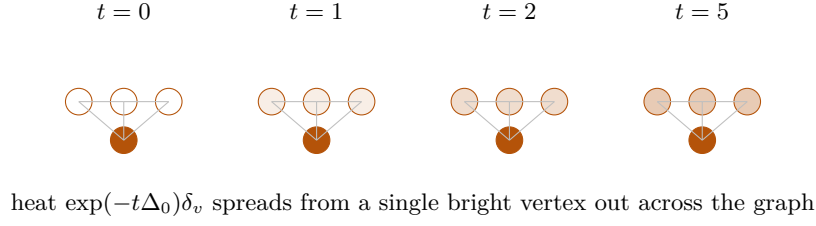
Architectural use. In GömbSoma the harmonic part of a vertex feature field captures features that survive a heat-flow smoothing (next section); these are the “topologically protected” features at the input scale.

5 Heat kernel

The heat equation on the simplicial complex is

$$\partial_t \omega_t = -\Delta_k \omega_t, \quad \omega_0 = f,$$

with solution $\omega_t = \exp(-t\Delta_k)f$. The operator $\exp(-t\Delta_k)$ is the *heat kernel*; it smooths features at spatial scale $\sqrt{t}$.



Architectural use. In GömbSoma the *Hodge term* in the Bochner-coupled message passing $(\alpha \cdot (\Delta_k h)_c)$ is one explicit step of heat diffusion; the layer learns to interpolate between flat connection (no smoothing), Hodge smoothing (continuous), and Ricci correction (geometric).

6 Forman-Ricci curvature

Definition 6.1 (Forman-Ricci $\kappa(e)$, unweighted). *For an edge $e = (u, v)$ in a simplicial complex,*

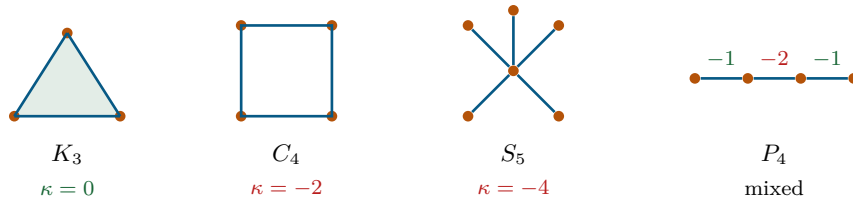
$$\kappa(e) = 2 - \deg(u) - \deg(v) + 2 \cdot |\Delta(u, v)|$$

where $|\Delta(u, v)|$ is the number of triangles incident on e .

Reading the formula.

- Each triangle adds $+2$ (a face filling the corner around e).
- Each degree adds -1 per incident neighbour (edge is being stretched between busy hubs).
- The leading $+2$ is the "flat-space" constant.

Complete graphs K_n achieve $\kappa = 0$ everywhere (triangle-balanced); cycles C_n ($n \geq 4$) achieve $\kappa = -2$ everywhere (no triangles, $\deg 2 + 2$). Stars S_n give strongly negative $\kappa = 1 - n$ (severe bottleneck).



The diagnostic. Strongly negative κ marks *bottlenecks*: edges over which information must pass but which are stretched thin. Topping et al. [2] showed that GNNs over graphs with strongly negative- κ edges exhibit *exponential* feature compression — the over-squashing problem. Their solution, SDRF (Section 9), iteratively rewires the graph to lift $\min_e \kappa(e)$.

Architectural use. In GömbSoma:

- Phase 2's **AdaptiveQuadtree** subdivides anchors with high $|\kappa_v|$ — pushing the bottleneck signal as a scale-refinement criterion.
- Phase 4's Bochner-coupled **HypergraphConv** adds a κ -weighted correction term to each message-passing step.
- Phase 6's SDRF rewiring adds shortcut edges where κ is too negative.

7 Ollivier-Ricci curvature (a contrast)

For comparison, Ollivier-Ricci κ_O is defined via optimal transport:

$$\kappa_O(u, v) = 1 - \frac{W_1(\mu_u, \mu_v)}{d(u, v)},$$

where μ_u is a probability measure on u 's neighbourhood (often uniform on $N(u)$) and W_1 is the 1-Wasserstein distance.

Why Forman instead. W_1 requires solving a linear-programming problem per edge — prohibitively expensive at the per-layer scale we need. Forman is purely combinatorial: degree counts and triangle counts, computable in $O(|E| \cdot \bar{d})$. Practically: Forman per layer, Ollivier reserved for post-hoc analysis if needed.

8 Bochner–Weitzenböck identity

In smooth Riemannian geometry, the Hodge Laplacian on 1-forms factors as

$$\Delta_1 \omega = \nabla^* \nabla \omega + \text{Ric}(\omega^\sharp)^\flat,$$

which decomposes the Laplacian into a *flat connection* term $\nabla^* \nabla$ (ordinary smoothing) and a *Ricci correction* term that bends the smoothing in directions of high curvature.

On a discrete simplicial complex the identity carries over with $\nabla^* \nabla$ replaced by the *vertex-incidence Laplacian* and Ric replaced by the Forman correction. This is exactly what we exploit in GömbSoma's Phase 4 Bochner-coupled HypergraphConv:

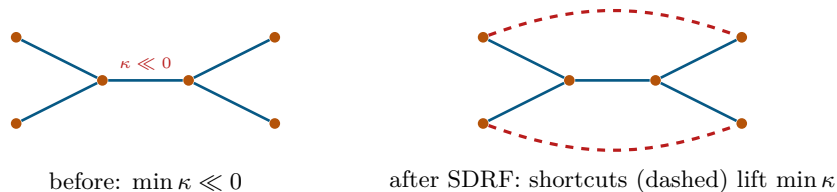
$$\text{msg}(c) = \psi_{\pi(c)} \left(\underbrace{\sum_i W_i x_{v_i}}_{\text{flat}} + \alpha \cdot \underbrace{(\Delta_k h)_c}_{\text{Hodge}} + \beta \cdot \underbrace{\sum_i \kappa(e_i) W_i x_{v_i}}_{\text{Ricci correction}} \right).$$

With $\alpha = \beta = 0$ the layer reproduces a vanilla message-pass. Increasing α adds heat-kernel smoothing; increasing β adds the curvature correction. The Bochner identity says these two terms are not arbitrary — they are the natural decomposition of the underlying geometric Laplacian.

9 Discrete Ricci flow and SDRF

Ricci flow on a smooth manifold deforms the metric in the direction of negative Ricci curvature: $\partial_t g_{ij} = -2\text{Ric}_{ij}$. Hamilton (1982) introduced it; Perelman used it to prove the Poincaré conjecture (2003).

Discrete analogue (SDRF, [2]). For each strongly-negative- κ edge in a graph, add a shortcut edge between unrelated neighbours of its endpoints. Iterate. The result is a graph with $\min_e \kappa(e)$ bounded away from $-\infty$ — *the over-squashing bottleneck has been relieved*.



Architectural use. Phase 6 of Ricci-Stim applies SDRF once to the multi-scale patch hypergraph before message passing; this is the principled answer to “our graph has a bottleneck through every patch in the 4-connected interior.”

10 Discrete Gauss–Bonnet (brief)

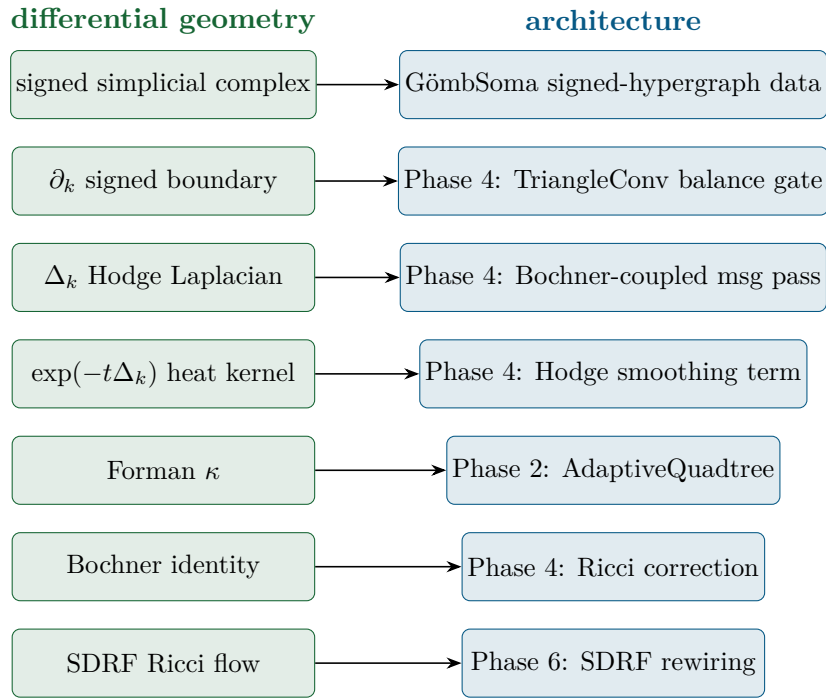
For a closed simply-connected planar region of a simplicial complex,

$$\sum_v K_v = \chi$$

where K_v is a suitable *vertex curvature* and χ is the Euler characteristic of the region. The full identity using Forman κ involves face contributions; on the unit tests we settle for the simpler combinatorial check $\sum_e \kappa(e)$ on small examples.

Architectural use. Sanity check that the Forman κ computation is correct (Phase 1 unit test).

11 Putting it all together — the architecture in one diagram



12 Further reading

References

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13 One-line takeaways

Concept	One-line takeaway for GömbSoma
Simplicial complex	vertices + edges + triangles + ...; the natural home of signed hypergraphs
Signed boundary ∂_k	alternating-sign map between simplex dimensions; the algebraic backbone of cycles
$\partial \circ \partial = 0$	boundary-of-boundary vanishes; makes the Hodge Laplacian well-defined
Hodge Laplacian Δ_k	per-dimension diffusion operator; harmonic kernel = topology-invariant features
Hodge decomposition	every feature = harmonic + gradient + curl
Heat kernel $\exp(-t\Delta_k)$	smoothing at spatial scale $\sqrt{t}$; the natural generalisation of Gaussian blur
Forman $\kappa(e)$	combinatorial Ricci: positive = clique, negative = bottleneck
Ollivier κ_O	optimal-transport Ricci; rigorous but expensive
Bochner–Weitzenböck	$\Delta_1 = \nabla^* \nabla + \text{Ric}$; the principled message-pass decomposition
SDRF Ricci flow	graph-rewiring to remove bottlenecks; addresses over-squashing
Discrete Gauss–Bonnet	$\sum \kappa = \chi$ on closed regions; a Forman sanity check

End of primer.